

Stats 141XP: Final Project

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March 8 2026

Abstract

This project analyzes subscriber and article data from The Pacific Divide to identify strategies for increasing audience engagement and growth. Using the Substack analytics data provided, we examine questions related to audience targeting, traffic sources, article title features, and optimal publishing time. Statistical methods including ANOVA, LASSO regression, and bootstrap resampling were used to evaluate patterns in reader behavior and article performance. The results suggest that readers acquired through Substack's internal network show higher engagement, while certain title characteristics and publishing times are associated with higher article view counts.

1 Introduction

The Pacific Divide is a Substack newsletter by Dr Russ Wilcox focused on artificial intelligence, geopolitics, and global technology competition, particularly between the United States and China. As subscribership continues to grow, understanding how readers discover the newsletter and what drives their engagement is important . The client for this project, Dr. Wilcox, is interested in identifying data driven strategies that can help broaden the newsletter’s audience while maintaining a high level of reader engagement.

Substack newsletters rely heavily on audience discovery, content, and timing to reach readers. However, it is often unclear which factors most strongly influence a reader’s level of engagement with content. Identifying these patterns can help prioritize marketing efforts, improve editorial decisions, and optimize publishing strategies.

This project analyzes subscriber and article level data from The Pacific Divide to better understand reader behavior and engagement. In particular, we examine how subscribers discover the publication, how article titles influence reader interest, and whether the timing of publication affects article view counts. The results of these analyses aim to provide realistic recommendations that can help with audience growth and interaction for the newsletter.

1.1 Objectives

The three research questions we want to answer with this project are:

1. Which traffic sources bring high-quality readers?
2. What title features most effectively drive audience engagement as measured by article views?
3. Which publishing days and hours are associated with higher article views?

2 Data

2.1 About the Data

The data used in this project was obtained from the Substack analytics export for The Pacific Divide. The dataset contains detailed information about subscriber activity, engagement metrics, subscription sources, and article performance.

The subscriber dataset includes a variety of behavioral metrics such as email opens, post views, comments, shares, and links clicked. Additional variables describe subscriber characteristics such as country, state or province, subscription type (free or paid), and the source through which the subscriber joined the publication. Several variables in the dataset capture how readers interact with the newsletter’s email distribution system. These include the number of emails received, opened, or dropped, as well as the number of unique emails and posts viewed by each subscriber. These variables help provide insight into how actively subscribers consume content.

In addition to the subscriber dataset, an articles dataset was used to analyze characteristics of individual posts. These variables include article titles, publishing dates and times, and total view counts. These variables are important for examining how title features and publication timing relate to audience engagement.

The project also uses a traffic sources dataset from Substack analytics, which records how readers arrive at the publication. Each row in this dataset represents a specific referral source, such as Email, LinkedIn, Google, or Direct traffic. For each source, the dataset reports the number of article views generated during the observation period as well as the number of subscribers driven by that source. Two variables in particular capture subscriber acquisition—free subs driven and paid subs driven—which represent the number of new free and paid subscribers attributed to each traffic source.

For the subscriber overview section (descriptive analysis), the subscriber dataset was used. To answer the first research question, the subscriber dataset was also used, while the second and third research questions were addressed using the articles dataset. A separate traffic sources dataset was used only for exploratory data analysis related to the first research question.

2.2 Data Preparation

Before conducting the statistical analyses, several data preparation steps were performed to transform the raw Substack data into formats suitable for analysis.

First, several variables required cleaning. For example, the variable Subscription Source (Free) in the subscriber dataset contained many inconsistently formatted entries. To make the data more interpretable, these raw sources were grouped into broader categories such as Direct, Substack Network, Substack Platform, Social, Article Page, Shared Link, and Other. We also noticed that one observation corresponded to Dr. Russ Wilcox himself (the client), so this observation was removed in order to avoid any bias and inconsistencies in the analysis.

Furthermore, additional feature engineering was performed for the article title analysis for the articles dataset. Several linguistic features were extracted from the article titles, including word count, character count, average word length, and the length of the longest word. Binary indicator variables were created to capture whether a title contained certain elements such as references to AI, China, or first-person language. Natural language processing techniques were also used to compute TF-IDF scores in order to quantify how distinctive specific words were relative to the full set of titles. These transformations helped convert the raw data into consistent variables that could be used in statistical models and analysis.

3 Subscriber Overview (Descriptive Analysis)

To provide context for the analyses that follow, we first summarize the basic characteristics of the subscriber base. The subscriber dataset contains 312 subscribers with varying levels of engagement with the newsletter. Subscribers exhibit a wide range of activity levels, from those who rarely interact with emails to those who regularly open and engage with content.

To better describe this variation in behavior, subscribers were grouped into engagement categories based on their email open rate in the past 30 days. These categories include inactive, low, medium, and high engagement levels. This grouping allows us to summarize how engagement is distributed across the audience and provides a foundation for later analysis of behavioral patterns.

In addition to engagement levels, we also examine the geographic distribution of subscribers. Mapping subscriber locations provides insight into where the audience is concentrated and helps identify major markets for the newsletter. Together, these descriptive summaries offer a general overview of the subscriber base before moving into deeper exploratory analysis of engagement patterns and content interactions.

3.1 Free vs Paid Subscribers

Table 1: Distribution of Free and Paid Subscribers

Subscriber Type	Count	Percentage
Free	300	96.15%
Paid	12	3.85%
Total	312	100%

The dataset consists primarily of free subscribers, with only a small portion of users upgrading to paid plans. As shown in Table 1, approximately 96.15% of subscribers are free users, while only 3.85% have subscribed to a paid plan. This distribution reflects a typical pattern in subscription-based newsletters, where a large base of free readers supports a much smaller group of paid subscribers. Understanding this imbalance provides important context for the subsequent analysis of engagement patterns and conversion behavior.

3.2 Basic Engagement Behaviour

Table 2: Summary of Subscriber Engagement Metrics

Engagement Metric	Mean Value
Emails Opened (30 Days)	5.26
Post Views (30 Days)	0.70
Comments	0.21
Shares	0.47

To further describe subscriber behavior, we summarize several basic engagement metrics that capture how readers interact with the newsletter content. These metrics include email opens, post views, comments, and shares. As shown in Table 2, subscribers demonstrate varying levels of activity across these measures. On average, subscribers open a modest number of emails and view several posts within a 30-day period, while active participation behaviors such as commenting and sharing occur less frequently. These descriptive statistics highlight the generally passive nature of most interactions, where readers primarily consume content rather than actively engage with it.

3.3 Engagement Groups

Table 3: Summary of Subscribers by Engagement Group

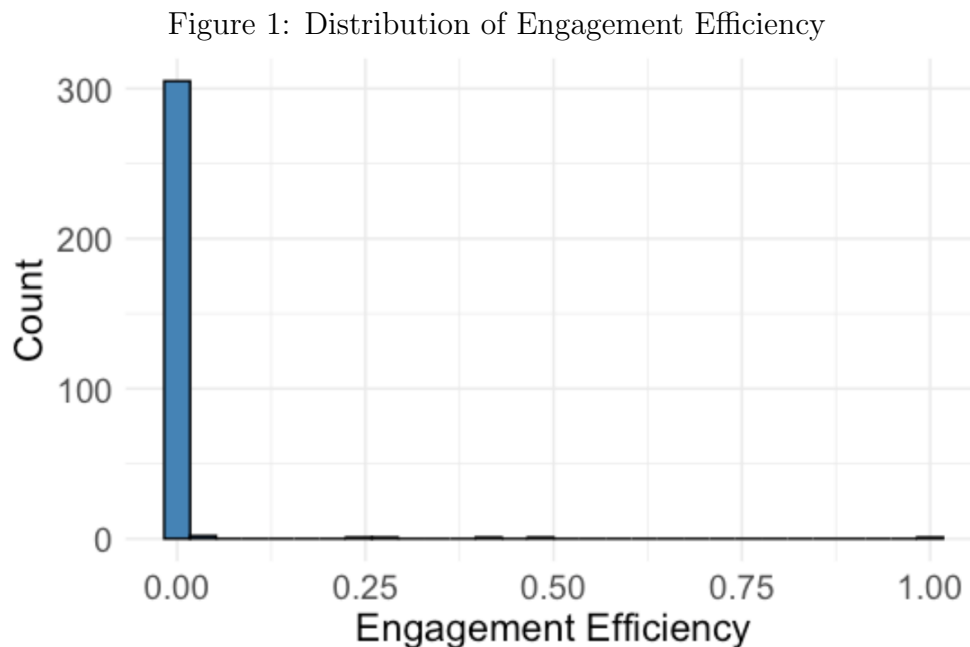
Engagement Group	Number of Subscribers
Inactive	133
Low	39
Medium	73
High	67

The table above summarizes the distribution of subscribers across engagement segments, which were defined based on the email open rate in the past 30 days. Subscribers with an open rate of 60 percent or higher were classified as highly engaged, those with open rates between 30 percent and 60 percent were categorized as moderately engaged, and those below 30 percent were labeled inactive.

As shown in the table, 67 subscribers are highly engaged, 73 are medium, 39 are low, and 133 are inactive. The distribution suggests a polarized audience structure, with a large core group consistently interacts with content, while a similarly large group exhibits low engagement. The bimodal structure of the distribution suggests that subscriber behavior clusters into highly active and minimally active groups rather than forming a smooth distribution.

3.4 Engagement Efficiency

Apart from using email open rate in the past 30 days as a measurement for engagement, we also created a new variable named “engagement efficiency”, defined as the number of comments and shares in the past 30 days divided by the number of emails opened in the past six months. It reflects whether those opens translate into active participation.



Our exploratory analysis above reveals that engagement efficiency is mostly 0 with the majority of subscribers exhibiting no commenting or sharing behavior, regardless of paid and non-paid users. The mean is 0.007982, which is approximately 0. This suggests that engagement is largely passive. Higher engagement efficiency values represent subscribers who actively interact with content relative to their reading frequency, suggesting a very small group of highly engaged readers. This distinction allows us to differentiate between passive readers and actively participating subscribers, which may be particularly relevant when identifying high-value or loyal users. The author may consider more ways of engaging readers in the articles.

3.5 Regional Distribution

Top Subscriber Markets and Paid Performance

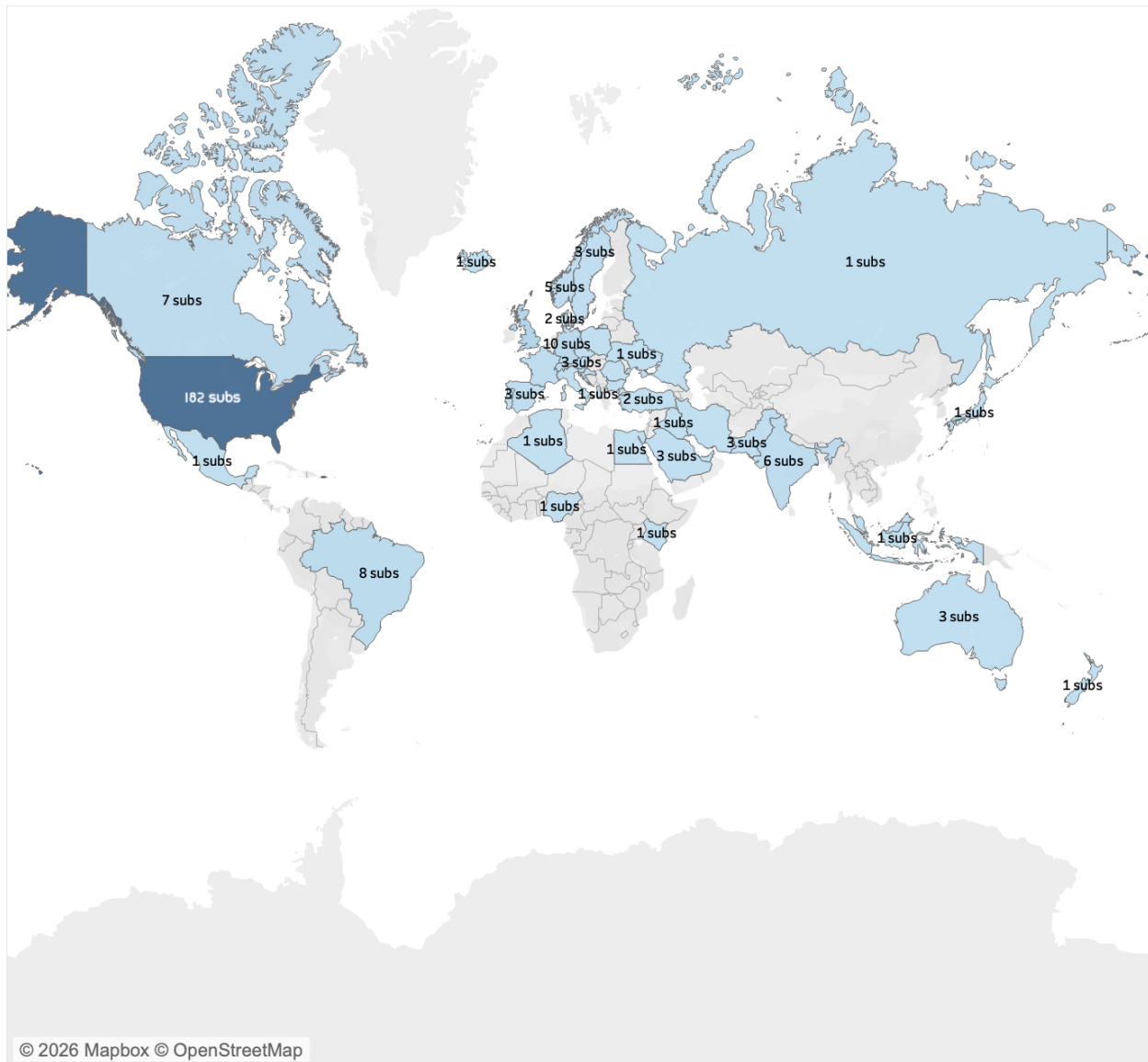


Figure 2: Top Subscriber Markets and Paid Performance

The map shows that subscribers are highly concentrated in the United States, which dominates both total subscribers (182) and revenue (\$1212). Most other countries have very small subscriber counts, typically fewer than 10, indicating a strong geographic imbalance.

Although a few smaller countries display relatively high paid rates (e.g., 33% or 50%), these are based on very small subscriber numbers and therefore may not be stable indicators of performance. Revenue largely mirrors subscriber volume rather than paid rate, suggesting that scale drives monetization more than conversion percentage.

Overall, the business is heavily dependent on the U.S. market, with international markets showing limited penetration and low revenue contribution.

3.6 Summary

Taken together, these descriptive statistics reveal several important structural features of the subscriber base. First, the audience is overwhelmingly composed of free subscribers, with only a small proportion converting to paid plans. Second, engagement is highly uneven. While many subscribers open emails and consume content, only a small subset actively interacts through commenting or sharing. Third, the subscriber base is geographically concentrated in the United States, suggesting that audience growth may depend heavily on domestic reach.

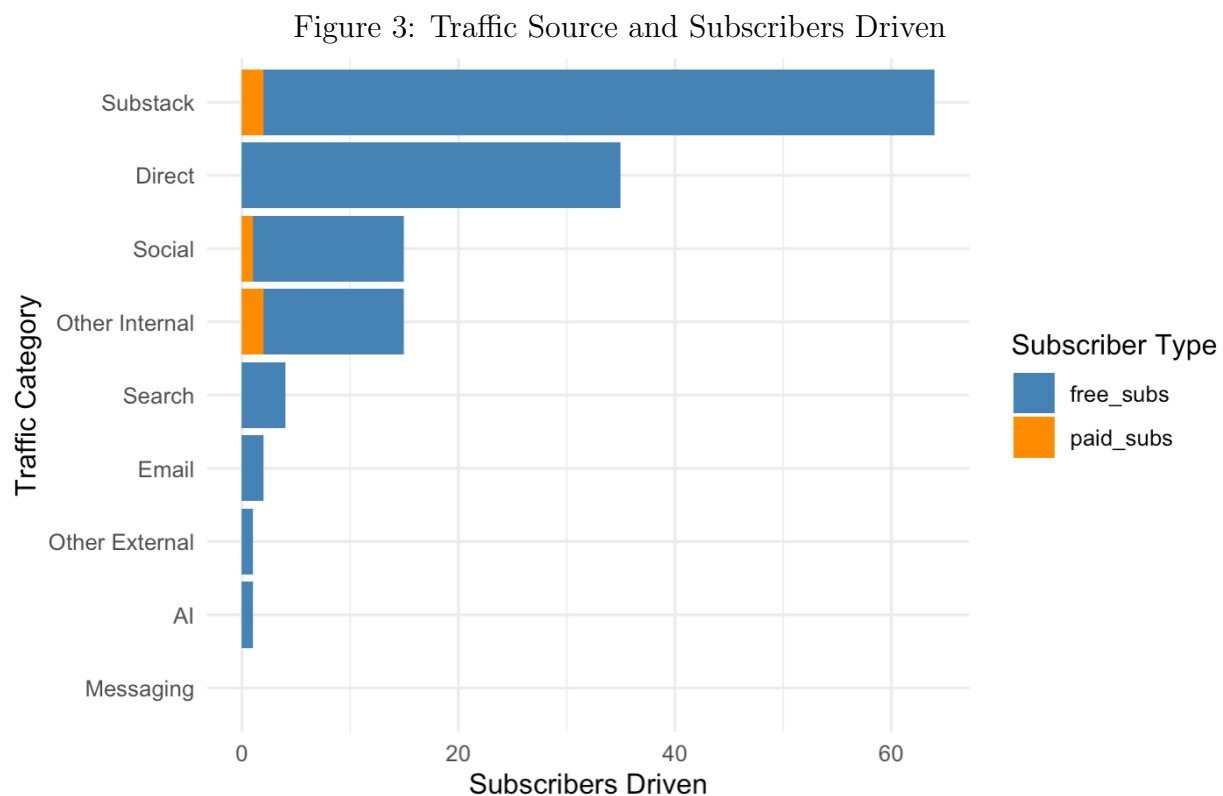
These patterns motivate several further questions explored in the following analysis. In particular, we investigate how subscriber acquisition channels relate to engagement behavior, whether linguistic features of article titles influence readership, and whether publication timing affects article visibility. By combining exploratory data analysis with statistical modeling, the following sections aim to identify factors that may contribute to audience growth and reader engagement.

4 Exploratory Data Analysis

4.1 Question 1 (traffic source)

Question: Which traffic sources bring high-quality readers?

To analyze how readers reach the publication, we first examine the traffic sources dataset as mentioned in the data section. This dataset records the referral source of article views and the number of free and paid subscribers generated from each source, allowing us to compare the effectiveness of different platforms in converting readers into subscribers.



The figure above presents the number of free and paid subscribers driven by each traffic source category. The results show that Substack's internal network generates the largest number of subscribers overall, with the majority being free subscribers and only a small number of paid conversions. Direct traffic also contributes a substantial number of new subscribers, indicating that readers who navigate directly to the publication are relatively likely to subscribe.

Table 4: Counts of Unique Values in Subscription Source Free

Subscription Source	Count
direct	36
substack-app	21
substack-reader-profile-discovery	20
linkedin.com	19
substack-notes	18
share	17
substack-signup-flow	14
post-page-subscribe-prompt	11
direct-to-app	10
post-end-cta	6
www.linkedin.com	5
m.facebook.com	4
import	3
substack.com	3
gift	2
l.facebook.com	2
www.google.com	2
bsky.app	1
com.reddit.frontpage	1
email-upsell	1
mail.google.com	1
open.substack.com	1
substack-follow-recommendations	1
substack-post-unlock	1
substack-your-weekly-stack	1
usc-word-edit.officeapps.live.com	1
www.noahpinion.blog	1

After constructing these broader traffic source categories, we shift our analysis to the subscriber dataset, which provides more detailed information on individual subscriber behavior and engagement. The variable Subscription Source Free contained a large number of highly specific source labels, many of which appeared only a small number of times. Analyzing the variable in its original form would therefore lead to sparse categories and unstable statistical estimates. To address this issue, the unique sources were systematically recategorized into a smaller set of broader traffic source groups. This recategorization served two primary purposes. First, it ensured that observations within each category represented conceptually similar types of traffic. Second, it produced a more balanced distribution of observations across categories, which improves the reliability of summary statistics and enables the use of statistical methods that require sufficiently sized groups.

4.2 Question 2 (article title)

Question: What title features most effectively drive audience engagement as measured by article views?

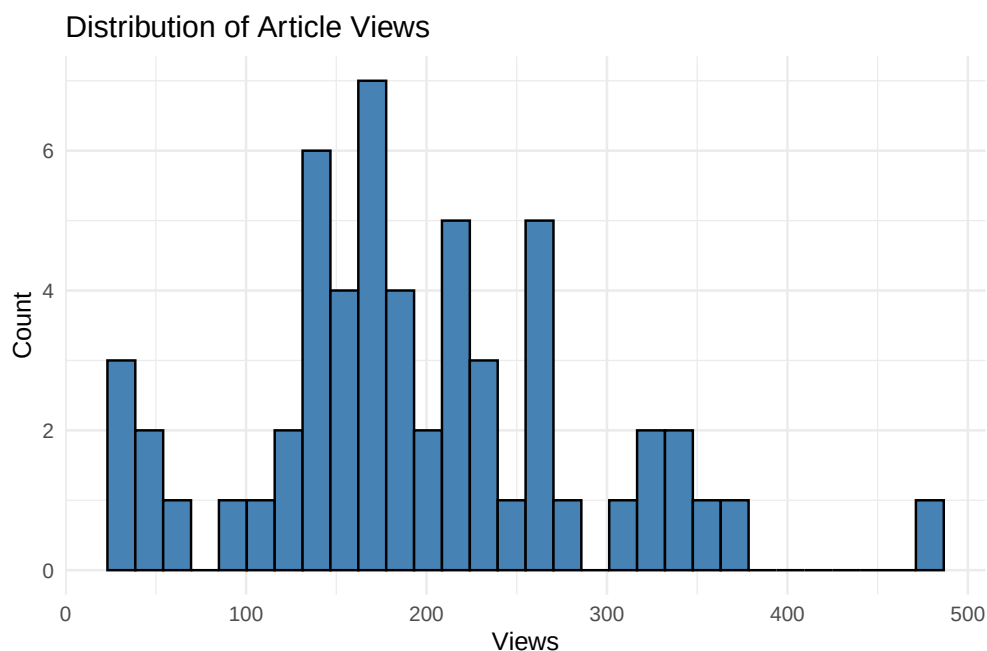


Figure 4: Distribution of the variable “views”

The distribution of article views appears reasonably well-behaved, with most observations concentrated between approximately 120 and 250 views and the highest density occurring around 150–200 views. While a small number of articles receive substantially higher view counts—extending beyond 300 views and approaching 500 in one case—the overall distribution does not appear excessively skewed. The majority of observations remain clustered within the central range, indicating a relatively stable distribution for analysis. Although a few high-performing articles contribute to a slight right tail, the distribution remains appropriate for use as the outcome variable in the methods applied later in the analysis.

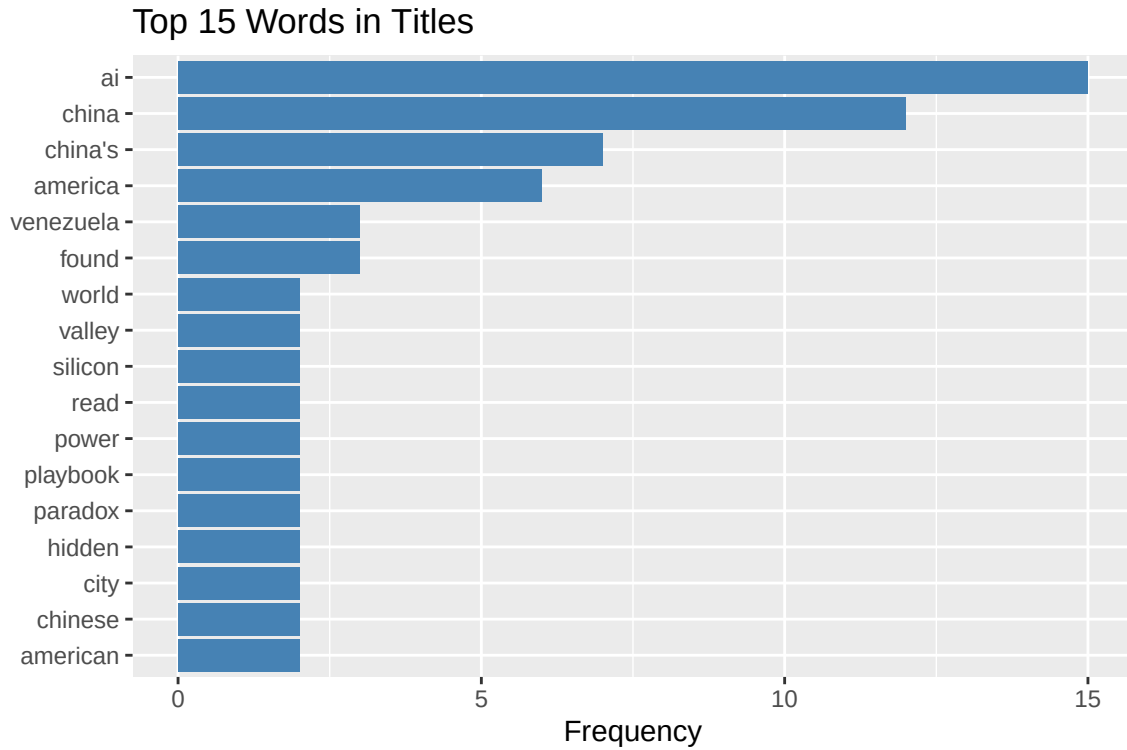


Figure 5: Top 15 Most Frequent Words in Titles

The most frequently occurring words are “AI”, appearing 15 times, and “China” or “Chinese”, appearing a total of 21 times. Beyond this, most words appear only once per document, indicating limited repetition across titles. This suggests that while AI and China are central themes in the dataset, the surrounding language varies considerably, likely reflecting different subtopics or perspectives within the broader subject area.

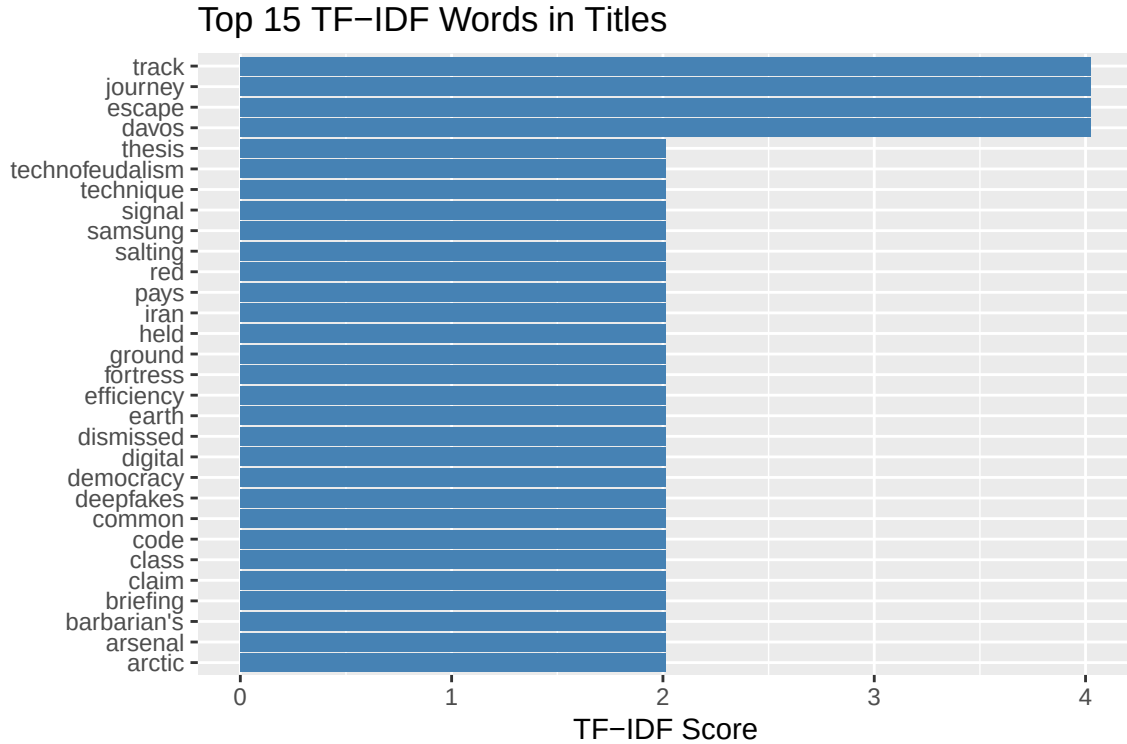


Figure 6: Top 15 Distinctive Terms (TF-IDF)

TF-IDF increases when a word appears frequently within a specific title but decreases when that word appears commonly across many titles. This allows us to capture distinctive or unique terms rather than simply frequent ones. Words such as “AI”, while frequent overall, may receive lower TF-IDF scores if they appear in many titles. The maximum TF-IDF (term frequency–inverse document frequency) score among words in the title. This measures the presence of a highly distinctive or uncommon word relative to the corpus. The graphs display the most distinctive terms.

4.3 Question 3 (optimal publish time)

Question: Which publishing days and hours are associated with higher article views?

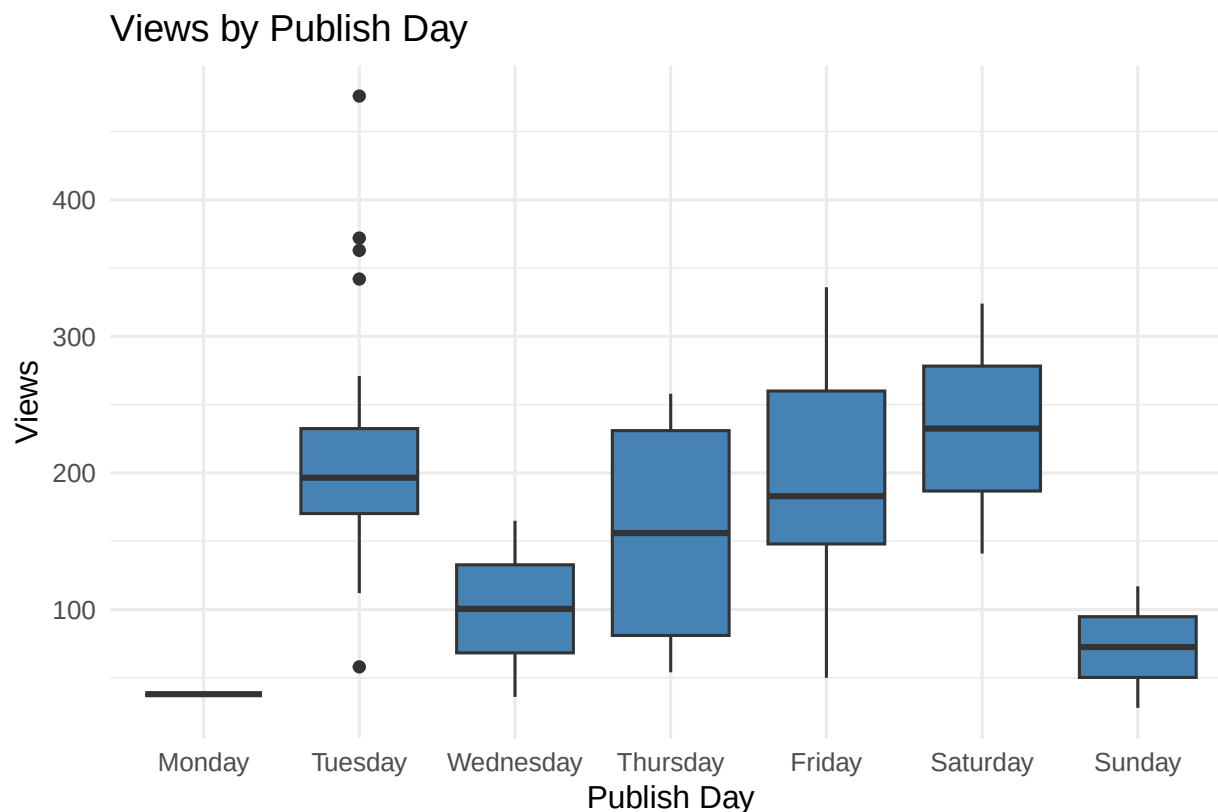


Figure 7: Article Views by Publish Day

To assess whether the relationship between article views and publishing day was worth further statistical testing, we first examined a boxplot of views by day of the week. The plot suggests that average article views vary across publishing days. In particular, Saturday appears to have the highest average number of views, while Sunday shows the lowest. This preliminary pattern indicates that publishing day may be associated with differences in article performance and motivates further statistical analysis to formally evaluate these differences.

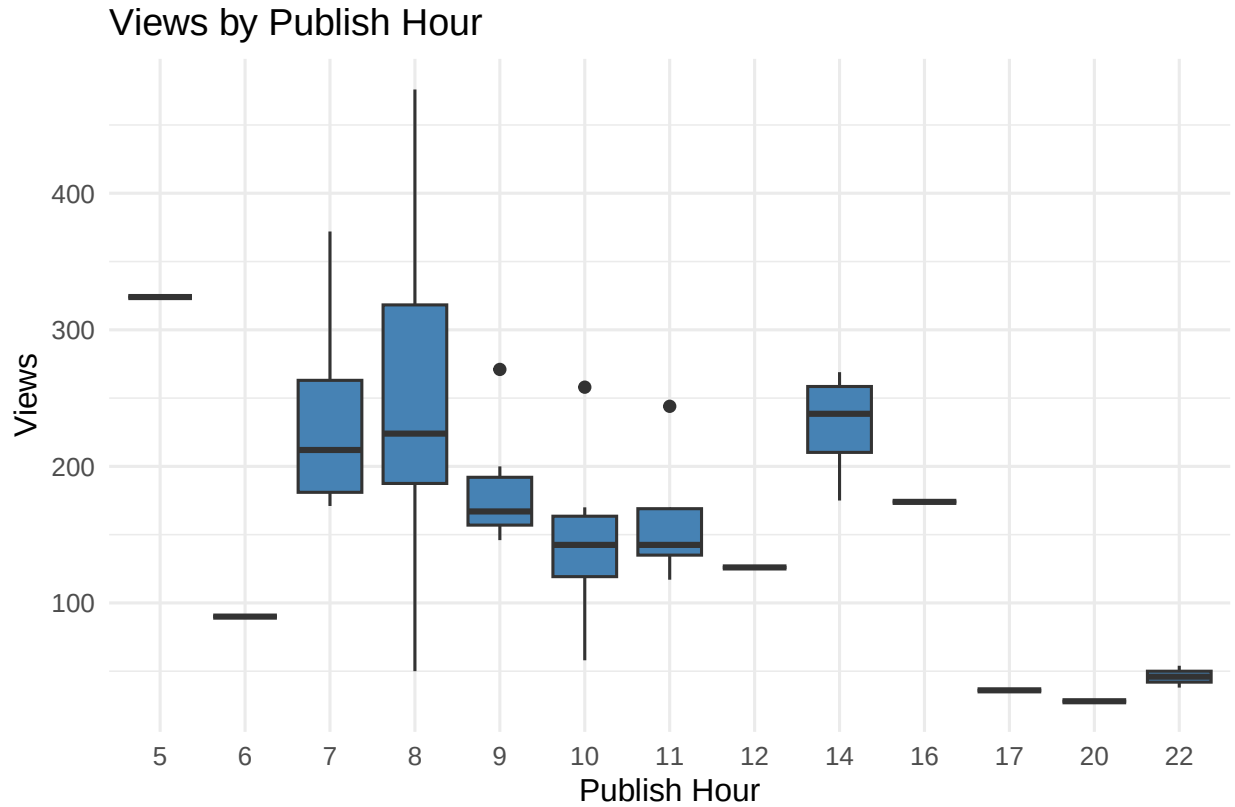


Figure 8: Article Views by Publish Hour

We conducted a similar exploratory analysis for the relationship between article views and publishing hour. The plots suggest that average article views also vary across publishing times. In particular, articles published at 2 PM appear to have the highest average number of views, while those published at 8 PM show the lowest. This pattern indicates that publishing time may also influence article performance and warrants further statistical investigation.

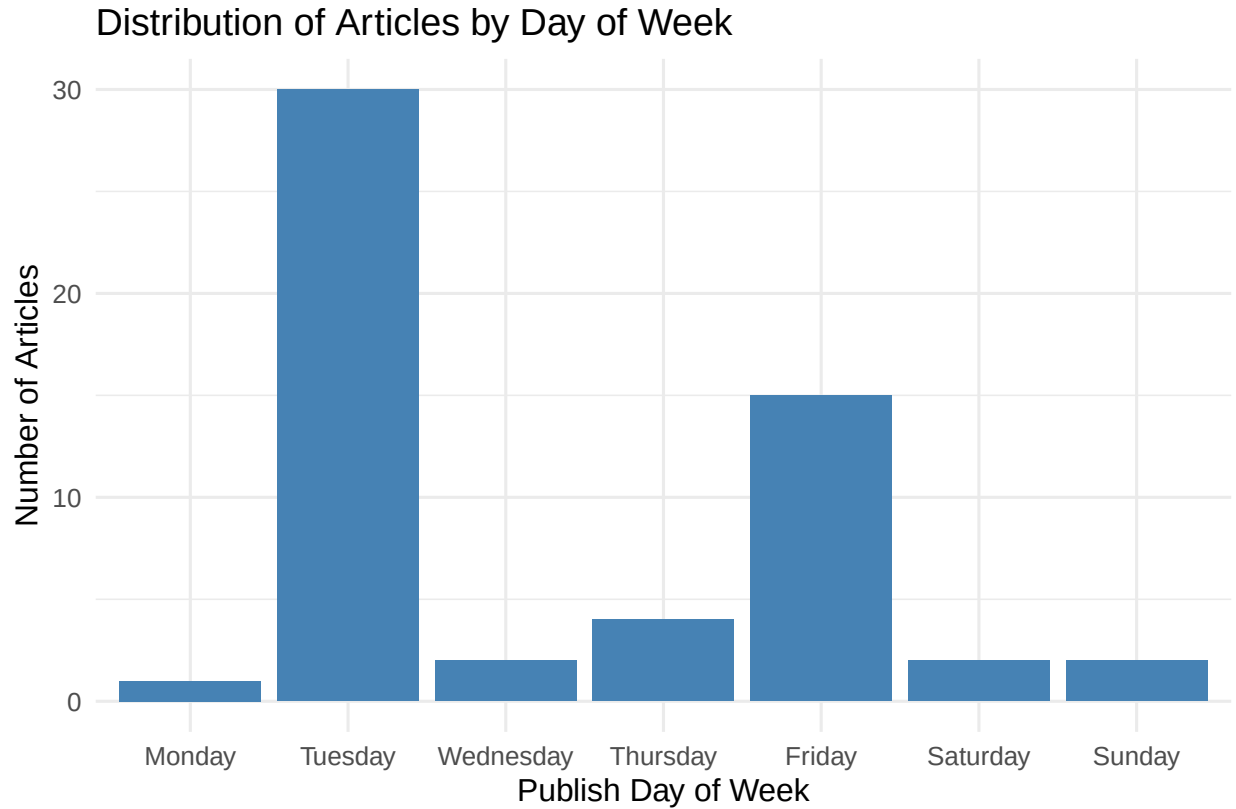


Figure 9: Article Count by Publish Day

The distribution of publishing days is highly uneven. Most articles were published on Tuesday, with Friday being the second most common day, while all other days contain very few observations. In particular, only four articles were published on Thursday and just one to two articles on the remaining days. This imbalance suggests that traditional parametric methods, such as ANOVA, may be less reliable due to highly unequal sample sizes across groups. As a result, a more appropriate approach is needed to test for differences in article views across publishing days.

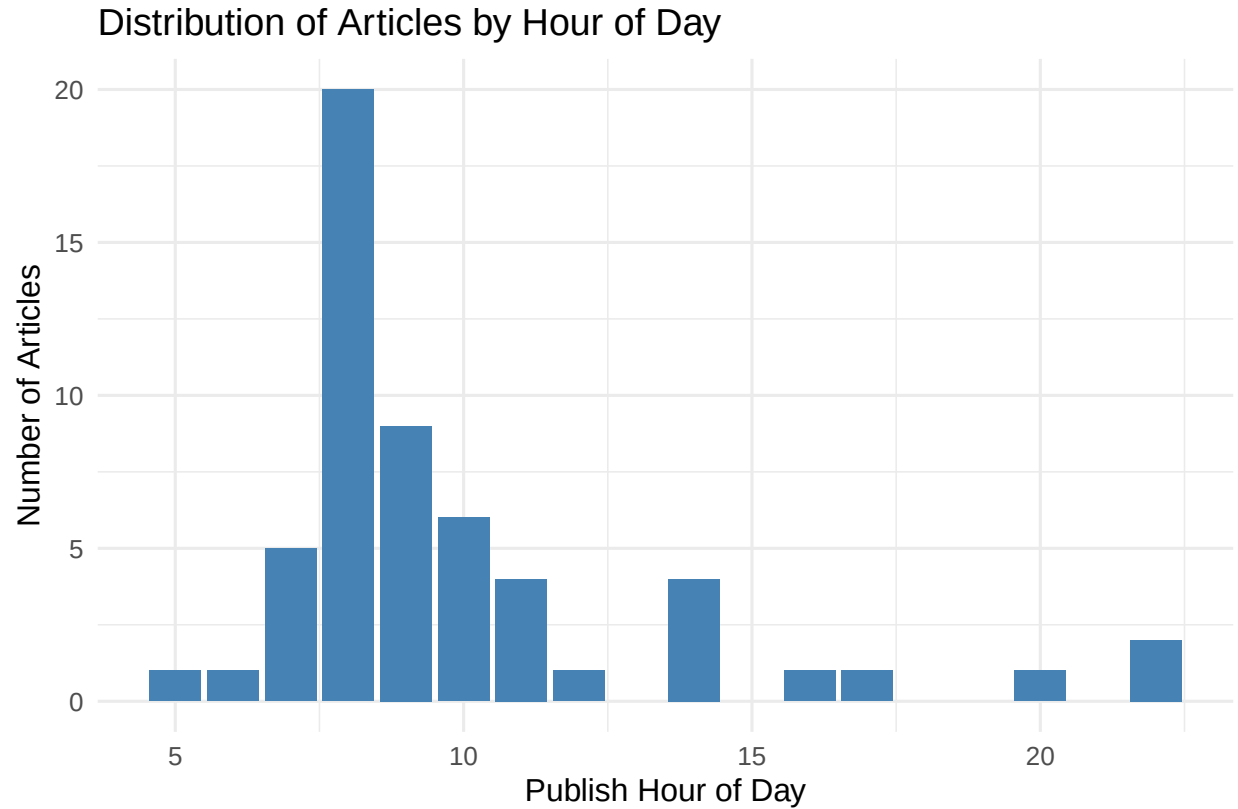


Figure 10: Article Count by Publish Hour

The same pattern is observed for publishing hour. The distribution is highly uneven, with the majority of articles being published at 8 AM, while many other hours contain only a small number of observations. This imbalance similarly limits the reliability of traditional parametric tests and further supports the use of more flexible approaches when analyzing differences in article views across publishing times.

5 Methods

5.1 Question 1 (traffic sources)

Question: Which traffic sources bring high-quality readers?

To begin, we focused on the variable Subscription Source Free, as the vast majority of subscribers in the dataset were free subscribers. However, the raw values in this variable contained many different and inconsistently structured sources, making direct analysis difficult. With the use of `grepl` function in the base R package, we developed a method to systematically categorize these sources into seven broader traffic source groups:

- **Direct:** Subscribers who accessed the publication directly, such as through manually entering the URL or using bookmarks. This category includes sources matching the pattern "direct".
- **Substack Network:** Subscribers who discovered the publication through social interactions within the Substack ecosystem, such as Notes, profile visits, or mentions by other Substack users. These sources reflect discovery through the networked, user-to-user features of the platform. This category includes sources matching the patterns "notes", "profile", or "mentions".
- **Substack Platform:** Subscribers who discovered the publication through Substack's general platform infrastructure rather than through social interactions with other users. This includes sources that contained the pattern "substack" but were not associated with Notes, profiles, or mentions. Because all Substack Network sources also contained the substring "substack", those observations were identified first, and the remaining matches were categorized as Substack Platform.
- **Social:** Subscribers arriving from external social media platforms such as LinkedIn, Facebook, Reddit, or Bluesky. This category includes sources matching the patterns "linkedin", "facebook", "reddit", or "bsky".
- **Article Page:** Subscribers who joined directly from an article page (e.g., post prompts or post-end calls to action). This category includes sources matching the pattern "post".
- **Shared Link:** Subscribers who joined through links shared by others, such as referral or forwarded links. This category includes sources matching the pattern "share".
- **Other:** Sources that did not clearly fall into the above categories and did not match any of the specified patterns. These included sources such as AI-related sites, gifted subscriptions, email sign-ups, and imported subscriber lists, all of which occurred only a small number of times in the dataset.

Using this newly created categorical variable and the total number of unique posts viewed by each subscriber, we conducted a one-way ANOVA to test whether the mean number of posts viewed differed across traffic source categories. In this analysis, subscriber engagement is operationalized as the number of unique posts viewed by each subscriber.

After observing a statistically significant ANOVA result, a Tukey post-hoc test was conducted to identify which specific traffic source groups differed in their mean engagement levels. This allowed us to determine which traffic sources are associated with higher levels of subscriber engagement. The results provide practical insight for the client by identifying which acquisition channels may be most effective for attracting higher-quality subscribers (or people that actually read the posts).

5.2 Question 2 (article title)

Question: What title features most effectively drive audience engagement as measured by article views?

Because the client emphasized an AI-first approach, we engineered additional title-based features using natural language processing techniques to better capture linguistic structure and thematic signals. The following variables were constructed for analysis:

- **Title_Word_Count:** The total number of words in the title. This measures title length and overall verbosity.
- **Title_Character_Count:** The total number of characters in the title. This provides a finer measure of length beyond word count.
- **Avg_Word_Length:** The average number of characters per word in the title. This serves as a proxy for linguistic complexity.
- **Longest_Word_Length:** The number of characters in the longest word in the title. This captures the presence of highly complex or technical terms.
- **Title_Has_Verb:** A binary indicator (1 = yes, 0 = no) for whether the title contains at least one verb. This measures whether the title uses action-oriented framing.
- **Title_Has_I:** A binary indicator for whether the title contains the word “I,” capturing first-person or personal narrative framing.
- **Title_Has_AI:** A binary indicator for whether the title references artificial intelligence (“AI”). This measures topic emphasis on AI-related content.
- **Title_Has_China:** A binary indicator for whether the title references China or Chinese-related topics. This captures potential geographic or geopolitical framing effects.

- **Title_TFIDF_Max:** The highest TF-IDF score among all words in a given article title. This value represents the single most distinctive or unique word in the title relative to the rest of the corpus. A higher value indicates that the title contains at least one word that appears rarely across other titles, making it more distinctive.

To reduce dimensionality and mitigate overfitting given the small sample size ($n = 56$), LASSO regression was first employed as an exploratory variable selection method. The regularization parameter λ was selected using 10-fold cross-validation, and the value minimizing the cross-validated error ($\lambda = \lambda_{\min}$) was used for model estimation. Predictors with non-zero coefficients were subsequently entered into an ordinary least squares regression model for statistical inference.

OLS was selected for final estimation because the primary objective of this analysis is interpretability rather than predictive optimization. Given the limited sample size, more complex machine learning models could introduce unnecessary variance and reduce transparency without meaningfully improving explanatory insight. Variance inflation factors were then examined to ensure model stability and assess potential multicollinearity.

5.3 Question 3 (optimal publish time)

Question: Which publishing days and hours are associated with higher article views?

The distribution of publishing days in the dataset is highly uneven as shown in the EDA, with Tuesday and Friday accounting for the largest number of observations while other days contain relatively few articles. This imbalance can lead to unstable estimates and reduced reliability when applying traditional parametric methods. To address this issue, bootstrap resampling with 10,000 iterations was used to estimate confidence intervals for mean views across publishing days and times. Simply comparing mean views across publishing hours using boxplots can be misleading because the number of observations within each hour is unbalanced and in some cases extremely small. Bootstrap resampling provides a more reliable approach by repeatedly sampling from the observed data and recalculating the mean views for each hour. This process generates an empirical sampling distribution, allowing us to estimate variability and construct confidence intervals without relying on strong parametric assumptions.

In addition to publishing day, the analysis also considered publishing time. The original time variable was recorded in clock format (e.g., 8:55 AM). To simplify the analysis, a `publish_hour` variable was created by rounding times down to the nearest hour, so an article published at 8:55 AM was assigned to hour 8. Under this convention, a time of 12:00 AM would correspond to hour 0. Because no articles in the dataset were published at that time, the value 0 does not appear among the observed publishing hours. The `publish_hour` variable was then used in the bootstrap procedure to estimate confidence intervals for mean views across different publishing hours.

6 Results and Interpretation

6.1 Question 1 (traffic sources)

Question: Which traffic sources bring high-quality readers?

After conducting a one-way ANOVA with the number of unique posts viewed as the outcome variable and the newly created traffic source category as the predictor, the following results were obtained:

Table 5: One-way ANOVA results testing differences in mean number of unique posts viewed across traffic source categories.

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Traffic Source	6	339.2	56.53	3.871	0.00113**
Residuals	196	2862.3	14.60		

This ANOVA table shows that there is a statistically significant difference in the mean number of unique posts viewed across the different traffic source categories ($F(6, 196) = 3.871$, $p = 0.00113$). This result indicates that the average level of subscriber engagement, measured by the number of unique posts viewed, varies depending on the source through which subscribers joined the publication.

Because the overall ANOVA test is significant, we reject the null hypothesis that all traffic source groups have the same mean engagement level. However, the ANOVA test alone does not identify which specific groups differ from one another. Therefore, the Tukey post-hoc test will show which traffic source categories have significantly different mean engagement levels:

Table 6: Tukey post-hoc pairwise comparisons of mean unique posts viewed across traffic source categories.

Comparison	Difference	Lower CI	Upper CI	Adj. p-value
Direct – Article Page	0.111	-3.054	3.276	0.9999
Other – Article Page	-0.389	-4.746	3.968	0.9999
Shared Link – Article Page	1.317	-2.533	5.167	0.9492
Social – Article Page	-0.295	-3.649	3.059	0.9999
Substack Network – Article Page	3.216	-0.041	6.474	0.0555
Substack Platform – Article Page	-0.047	-3.267	3.172	1.0000
Other – Direct	-0.500	-4.321	3.321	0.9997
Shared Link – Direct	1.206	-2.026	4.437	0.9239
Social – Direct	-0.406	-3.027	2.215	0.9993
Substack Network – Direct	3.105	0.610	5.601	0.0050
Substack Platform – Direct	-0.159	-2.604	2.287	1.0000
Shared Link – Other	1.706	-2.700	6.111	0.9103
Social – Other	0.094	-3.885	4.073	1.0000
Substack Network – Other	3.605	-0.293	7.503	0.0904
Substack Platform – Other	0.341	-3.524	4.207	0.9999
Social – Shared Link	-1.612	-5.029	1.805	0.7983
Substack Network – Shared Link	1.899	-1.423	5.221	0.6146
Substack Platform – Shared Link	-1.364	-4.649	1.920	0.8785
Substack Network – Social	3.512	0.780	6.243	0.0032
Substack Platform – Social	0.248	-2.438	2.933	0.9999
Substack Platform – Substack Network	-3.264	-5.827	-0.700	0.0037

The post-hoc analysis reveals differences between several traffic source groups. In particular, statistically significant differences are observed between Substack Network and Direct ($p = 0.005$), Substack Network and Social ($p = 0.003$), and Substack Platform and Substack Network ($p = 0.004$). Additionally, the comparison between Substack Network and Article Page ($p = 0.055$) approaches statistical significance and may suggest a potential difference in engagement between these groups.

The confidence intervals from the Tukey post-hoc analysis provide additional insight into the magnitude and direction of the differences in engagement between traffic sources.

For the comparison between Substack Network and Direct traffic, the 95% confidence interval ranges from 0.61 to 5.60. Because the entire interval is positive and does not include zero, this indicates that subscribers who joined through the Substack Network view significantly more posts on average than those who joined through Direct sources. The interval suggests that Substack Network subscribers view approximately 0.6 to 5.6 more unique posts on average.

Similarly, the comparison between Substack Network and Social traffic yields a confidence interval from 0.78 to 6.24. Since this interval also lies entirely above zero, it indicates that Substack Network subscribers engage with significantly more posts than those arriving

through Social sources, with an estimated difference of roughly 0.8 to 6.2 additional unique posts viewed on average.

The comparison between Substack Platform and Substack Network has a confidence interval from -5.83 to -0.70. Because the interval is entirely negative, this indicates that subscribers from the Substack Network view significantly more posts than those from the general Substack Platform category. The interval suggests that Substack Network subscribers view approximately 0.7 to 5.8 more unique posts on average.

Finally, the comparison between Substack Network and Article Page has a confidence interval ranging from -0.04 to 6.47. Because this interval slightly includes zero, the difference is not statistically significant at the 0.05 level which is shown in the adjusted p-value of 0.0555. However, because the interval is largely positive, the results may suggest a potential tendency for Substack Network subscribers to view more unique posts than those joining directly from article pages, although the evidence is not strong enough to draw a definitive conclusion.

6.2 Question 2 (article title)

Question: What title features most effectively drive audience engagement as measured by article views?

These were the results for variable selection after running LASSO:

Table 7: LASSO Regression Coefficients for Article Title Features

Variable	Coefficient
Intercept	224.016
Title_Word_Count	0.000
Title_Character_Count	0.000
Avg_Word_Length	0.000
Longest_Word_Length	-11.084
Title_Has_Verb	59.095
Title_Has_I	10.359
Title_Has_AI	-26.528
Title_Has_China	48.317
Title_TFIDF_Max	12.819

As shown in the table, LASSO shrank several coefficients exactly to zero, effectively removing Title_Word_Count, Title_Character_Count, and Avg_Word_Length from the model. The remaining variables with non-zero coefficients were Longest_Word_Length, Title_Has_Verb, Title_Has_I, Title_Has_AI, Title_Has_China, and Title_TFIDF_Max.

However, because LASSO coefficients are penalized and primarily used for variable selec-

tion rather than statistical inference, the selected predictors were subsequently entered into an ordinary least squares regression model to estimate interpretable effect sizes and assess statistical significance.

The results for the fitted OLS model with the newly selected variables are the following:

Table 8: OLS Regression Results: Article Title Features and Views

Variable	Estimate	Std. Error	t-value	p-value
Intercept	210.155	65.919	3.188	0.0025 **
Longest_Word_Length	-12.144	5.489	-2.212	0.0316 *
Title_Has_Verb	70.336	27.716	2.538	0.0144 *
Title_Has_I	23.508	43.022	0.546	0.5873
Title_Has_AI	-37.018	26.603	-1.392	0.1704
Title_Has_China	62.289	27.994	2.225	0.0307 *
Title_TFIDF_Max	21.662	15.950	1.358	0.1806
Residual Standard Error: 79.4 (df = 49)				
Multiple R^2 : 0.326 Adjusted R^2 : 0.243				
F-statistic: 3.942 (6, 49), $p = 0.0027$				

To assess potential multicollinearity among the selected predictors, variance inflation factors (VIFs) were calculated:

Table 9: Variance Inflation Factors (VIF) for OLS Predictors

Variable	VIF
Longest_Word_Length	1.285
Title_Has_Verb	1.567
Title_Has_I	1.091
Title_Has_AI	1.233
Title_Has_China	1.632
Title_TFIDF_Max	1.814

As shown in the table, all VIF values were well below commonly accepted thresholds, indicating no evidence of problematic multicollinearity.

From the OLS results, we observe that several title characteristics are statistically associated with variation in article view counts. In particular, Title_Has_Verb is positive and statistically significant ($p = 0.014$), indicating that titles containing at least one verb are associated with approximately 70 additional views on average, holding other factors constant.

Similarly, Title_Has_China is positive and statistically significant ($p = 0.031$), suggesting that titles referencing China are associated with roughly 62 additional views.

In contrast, Longest_Word_Length is negative and statistically significant ($p = 0.032$),

implying that increased linguistic complexity—measured by the length of the longest word in the title—is associated with lower view counts. Specifically, each additional character in the longest word corresponds to an estimated decrease of approximately 12 views. This finding is also broadly consistent with prior research showing that shorter, clearer, and more informative titles tend to receive greater academic attention and citation impact (Paiva et al., 2012). More broadly, prior work has shown that linguistic characteristics of headlines can influence headline success and reader response (Gligorić et al., 2023). In addition, large-scale field experiments suggest that headline wording can influence click-through behavior, with certain forms of negative language associated with higher click rates (Robertson et al., 2023).

The model explains approximately 24.3% of the variation in article views (Adjusted $R^2 = 0.243$). Although a substantial portion of variability remains unexplained, this is expected given the multifaceted nature of audience engagement. Article views are shaped by numerous external factors beyond title construction, including subscriber growth over time, email open rates, platform algorithms, publication timing, and external traffic sources. Because this analysis isolates title-based linguistic features alone, explaining nearly one quarter of the variance represents meaningful explanatory power within the scope of the variables considered. The statistically significant F-test further indicates that, collectively, the selected title features provide information about variation in views beyond random noise.

The remaining variables, including Title_Has_I, Title_Has_AI, and Title_TFIDF_Max, were not statistically significant at conventional levels, indicating insufficient evidence that these features meaningfully influence article views in this sample.

6.3 Question 3 (optimal publish time)

Question: Which publishing days and hours are associated with higher article views?

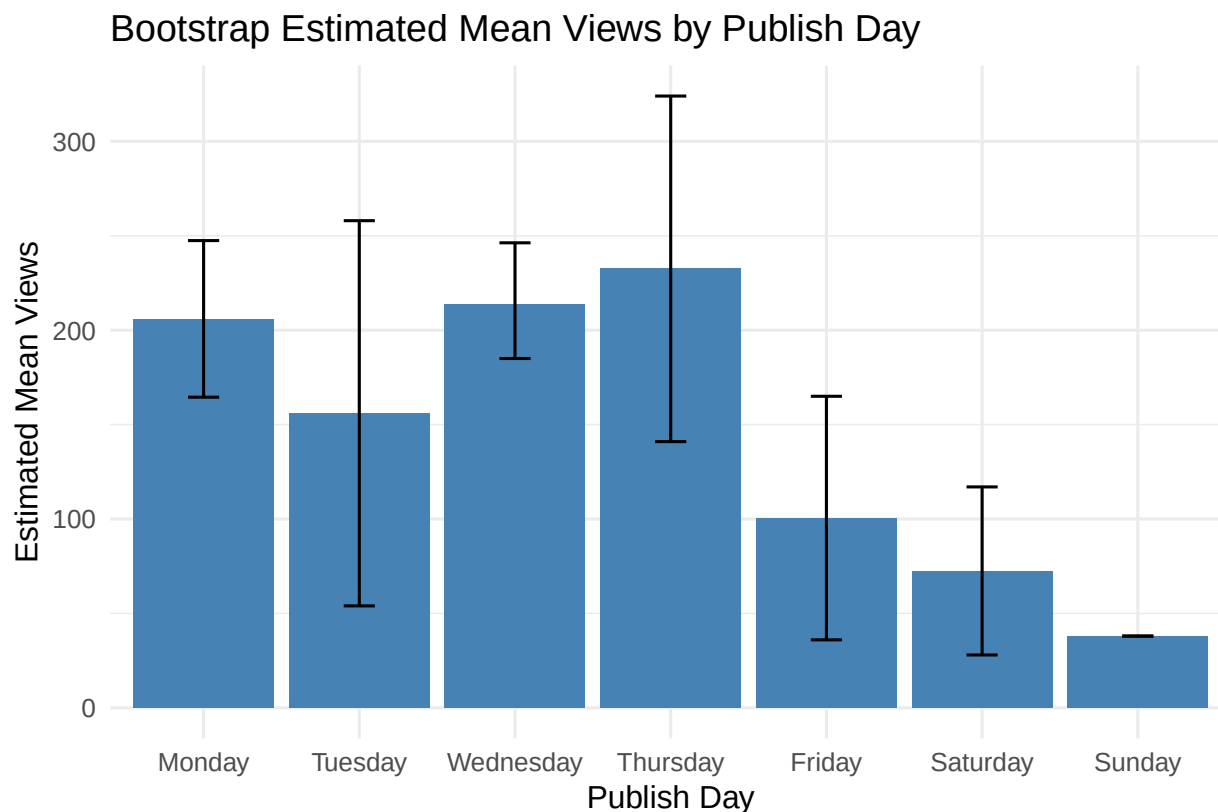


Figure 11: Bootstrap estimated mean article views by publishing day.

The bootstrap results indicate that articles published on Thursday have the highest estimated mean views, followed by Wednesday. This differs from the pattern observed in the earlier EDA boxplots, where Saturday appeared to have the highest views and Tuesday followed next. This difference likely reflects the instability of raw averages in the EDA due to very small sample sizes on certain days. The reason that Sunday shows no confidence interval is because only one article was published on that day. With only a single observation, bootstrap resampling repeatedly selects the same value, resulting in no variability and therefore a confidence interval of length zero. As a result, the estimate for Sunday is not reliable and should not be interpreted when comparing publishing days. The following table shows the exact bootstrap estimates of mean article views by publishing day.

Table 10: Bootstrap estimates of mean article views by publishing day with 95% confidence intervals.

Publish Day	Mean Views	Lower CI	Upper CI
Monday	205.77	164.53	247.45
Tuesday	156.03	54.00	258.00
Wednesday	213.71	185.00	246.28
Thursday	232.77	141.00	324.00
Friday	100.24	36.00	165.00
Saturday	72.20	28.00	117.00
Sunday	38.00	38.00	38.00

The table results show that articles published on Thursday have the highest estimated mean number of views (232.77), followed closely by Wednesday (213.71) and Monday (205.77). In contrast, articles published later in the week tend to receive fewer views on average, with Friday and Saturday showing substantially lower mean views.

Although Thursday has the highest estimated mean views, the substantial overlap among the confidence intervals for Monday, Wednesday, and Thursday suggests that the true differences between these days may not be statistically meaningful. Thursday has the widest interval (141.00 to 324.00), while Tuesday has the next widest interval (54.00 to 258.00). These wide intervals indicate greater variability in article views for these publishing days, resulting in less precise estimates of the mean.

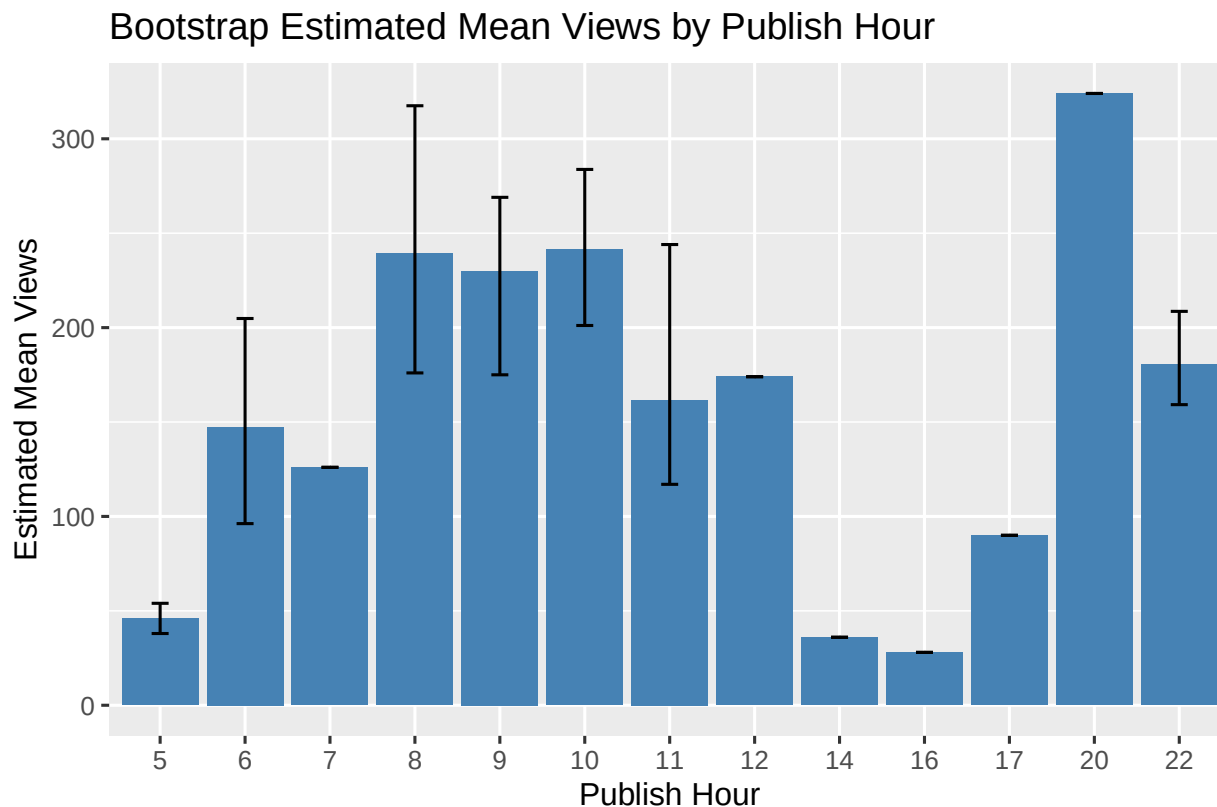


Figure 12: Bootstrap estimated mean article views by publishing hour.

The same procedure was applied to publishing hour. In the earlier exploratory data analysis, the boxplots suggested that articles published at 2 PM had the highest view counts. However, after applying the bootstrap procedure to estimate mean views and confidence intervals, articles published at 10 AM appear to have the highest estimated average views, followed by those published at 8 AM. Although 8 PM shows the highest mean value, it does not display a confidence interval because it is based on a single observation. Therefore, this result is not interpreted as meaningful evidence of higher performance at that time. The following table shows the exact bootstrap estimates.

Table 11: Bootstrap estimates of mean article views by publishing hour with 95% confidence intervals.

Publish Hour	Mean Views	Lower CI	Upper CI
5	45.99	38.00	54.00
6	147.34	96.20	204.83
7	126.00	126.00	126.00
8	239.31	176.00	317.50
9	230.10	175.00	269.00
10	241.23	201.13	283.77
11	161.76	117.00	244.00
12	174.00	174.00	174.00
14	36.00	36.00	36.00
16	28.00	28.00	28.00
17	90.00	90.00	90.00
20	324.00	324.00	324.00
22	180.47	159.20	208.60

Looking at the table, only the highlighted rows correspond to publishing hours with a sufficient number of observations to produce meaningful bootstrap confidence intervals. Among these hours, the highest estimated mean views occur at 10 AM (241.23), followed closely by 8 AM (239.31) and 9 AM (230.10).

The widest confidence intervals are observed for 6 AM (96.20 to 204.83) and 11 AM (117.00 to 244.00), indicating greater variability in article views for posts published during these hours. In contrast, the confidence interval for 10 AM (201.13 to 283.77) is relatively narrower while still maintaining a high estimated mean, suggesting a more stable estimate of article performance during that time.

Similar to the bootstrap results for publishing day, several confidence intervals overlap, particularly among 8 AM (176.00 to 317.50), 9 AM (175.00 to 269.00), 10 AM (201.13 to 283.77), and 11 AM (117.00 to 244.00). This overlap indicates that the true mean views for these publishing hours may not differ substantially. Overall, while certain publishing hours such as 8–10 AM show higher estimated mean views, the overlapping confidence intervals suggest that differences across many hours remain uncertain.

7 Conclusion

7.1 Question 1 (traffic source)

Question: Which traffic sources bring high-quality readers?

The goal of this analysis was to determine which traffic sources bring the most engaged readers to The Pacific Divide. In this analysis, engagement was measured by the number of unique posts viewed by each subscriber. This metric was selected because it reflects the extent to which a reader actively explores and consumes content across the publication rather than viewing only a single article. Subscribers who view multiple unique posts are more likely to interact with the publication over time, making this measure a useful indicator of “high-quality” subscribers.

The ANOVA results indicate that at least one traffic source category has a different mean engagement level compared to the others. Subsequent Tukey post-hoc comparisons further reveal that subscribers who discovered the publication through the Substack Network viewed significantly more posts on average than subscribers arriving through Direct traffic, Social media platforms, and the general Substack platform. Notably, all statistically significant pairwise differences involved the Substack Network category, indicating that this group demonstrates higher engagement relative to several other traffic sources.

These findings suggest that participation within the Substack networking ecosystem may be an effective way to attract highly engaged readers. Activities such as collaborating with other Substack writers or increasing participation in Substack’s internal social features may help expand audience reach and attract readers who actively consume the content.

While external social media channels remain useful for increasing visibility and attracting new subscribers, the results suggest that the most engaged readers are more likely to emerge from within Substack’s own discovery and networking mechanisms.

7.2 Question 2 (article title)

Question: What title features most effectively drive audience engagement as measured by article views?

This analysis examines whether specific structural and thematic characteristics of article titles are associated with variation in view counts. After applying LASSO for exploratory variable selection and ordinary least squares regression for statistical inference, several clear patterns emerged.

Titles containing verbs were associated with substantially higher view counts, suggesting that action-oriented framing may increase engagement. Titles referencing China were also

positively associated with views, indicating that topic emphasis may meaningfully influence audience interest. In contrast, greater linguistic complexity—measured by the length of the longest word in the title—was associated with lower view counts, supporting the idea that clarity and simplicity enhance accessibility.

Other engineered features, including first-person framing, AI references, and lexical distinctiveness, were not statistically significant in this sample. In particular, the lack of a significant effect for AI-related titles may reflect the context of the blog itself, where readers already expect AI-focused content. As such, explicitly referencing AI in the title may not provide additional signaling value. Although first-person framing and distinctive terminology were not statistically significant predictors in this model, experimenting with these stylistic variations may still be worthwhile from a strategic perspective. The absence of statistical significance does not necessarily imply the absence of practical value, especially given the modest sample size.

It is important to note that these findings represent statistical associations rather than causal relationships. Because this analysis is based on observational data rather than a controlled experiment, we cannot conclude that modifying a specific title feature will directly cause an increase or decrease in views. Instead, the results identify patterns within this dataset that may inform strategic experimentation moving forward.

7.3 Question 3 (optimal publish time)

Question: Which publishing days and hours are associated with higher article views?

Due to the variability reflected in the confidence intervals, it is not possible to identify a single publishing day that can be guaranteed to produce the highest number of views. The overlap among several intervals suggests that the differences between specific days may not be statistically meaningful. However, the general pattern in the bootstrap estimates indicates that articles published earlier in the week—particularly from Monday through Thursday—tend to receive higher average views than those published later in the week from Friday through Sunday.

Given that the client most frequently publishes on Tuesdays and Fridays, the results suggest that Tuesday may be the more favorable option between the two. While this does not guarantee higher performance for every article, publishing earlier in the week aligns more closely with the engagement patterns observed in the dataset.

A similar pattern is observed for publishing hour. Several hours could not be meaningfully evaluated because they contain too few observations to produce reliable confidence intervals. Among the hours with sufficient data, the estimates suggest that articles published in the morning—particularly between approximately 8 AM and 10 AM (or up to 11 AM when considering that times were rounded down to the nearest hour)—tend to receive the highest

average views. Since the client already publishes most articles around 8 AM, this timing appears to align well with the engagement patterns observed in the data.

Because the dataset contains a relatively small number of observations, even a modest increase in the number of articles would improve the precision and reliability of the bootstrap estimates. With more data across different publishing times, clearer patterns may emerge and allow for more confident recommendations regarding optimal publishing strategies. Overall, while the exact optimal publishing time cannot be determined with certainty from the current dataset, the results suggest that publishing earlier in the week during the morning hours provides a reasonable strategy for maximizing expected article views.

8 Limitations

Several limitations should be considered when interpreting the results of this analysis.

First, the dataset contains a relatively small number of observations for certain variables, particularly the article level dataset used to examine title features and publishing times. With only 56 articles available for analysis, statistical power is limited.

Second, the analysis identifies statistical associations between variables but cannot establish causal relationships. For example, while certain title characteristics are associated with higher view counts, these findings are supported only by statistical evidence and should not be interpreted as causal effects.

Another limitation arises from the uneven distribution of observations across categories. Some publishing days and hours contain very few articles, which may lead to biased estimates of average performance. Although bootstrap resampling was used to address this issue in the publishing time analysis, estimates for sparsely represented categories should still be treated with caution.

Additionally, subscriber engagement metrics may be influenced by nuances not captured in the dataset. Things that are not included in the data set like subscriber growth over time, email deliverability issues, social media exposure, and broader trends in reader interest could all impact the relationships analyzed. Because these variables are not observed in the data, they may contribute to unexplained variation in engagement outcomes.

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